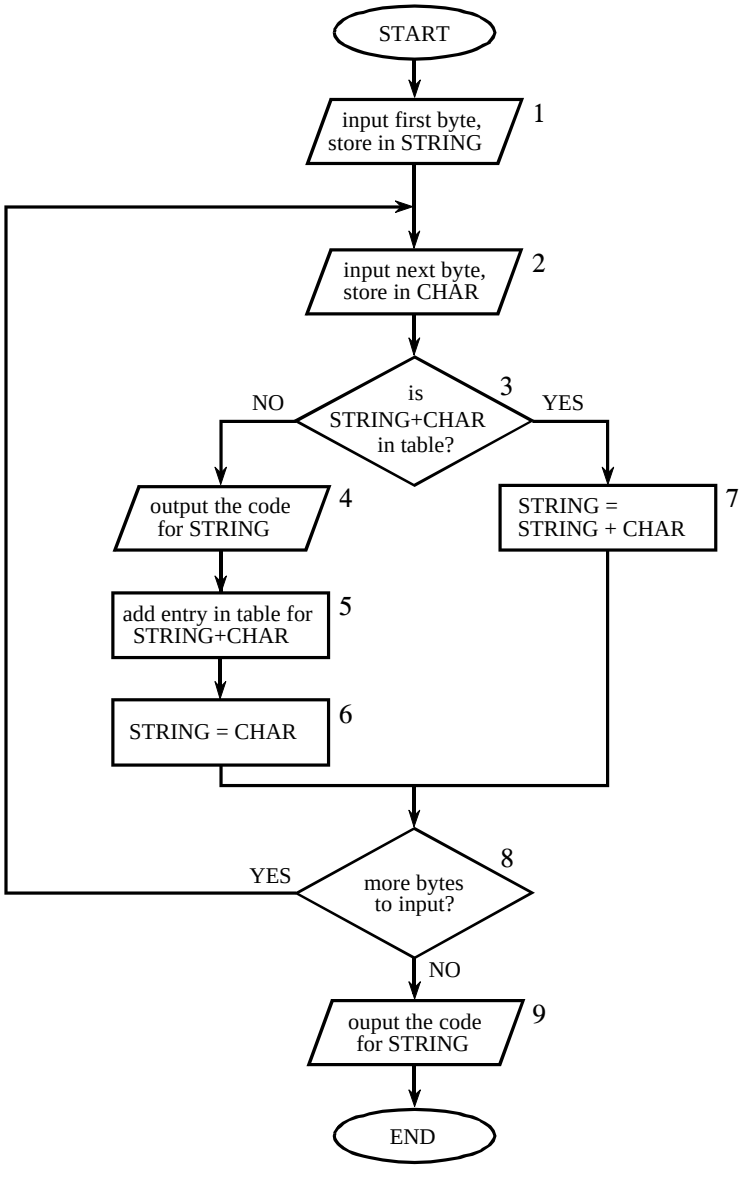

FIGURE 27-7
LZW compression flowchart. The variable, *CHAR*, is a single byte. The variable, *STRING*, is a variable length sequence of bytes. Data are read from the input file (box 1 & 2) as single bytes, and written to the compressed file (box 4) as 12 bit codes. Table 27-3 shows an example of this algorithm.

Figure 27-7 shows a flowchart for LZW compression. Table 27-3 provides the step-by-step details for an example input file consisting of 45 bytes, the ASCII text string: *the/rain/in/Spain/falls/mainly/on/the/plain*. When we say that the LZW algorithm reads the character "a" from the input file, we mean it reads the value: 01100001 (97 expressed in 8 bits), where 97 is "a" in ASCII. When we say it writes the character "a" to the encoded file, we mean it writes: 000001100001 (97 expressed in 12 bits).